THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 7: **Work, Energy and Simple Machines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The mechanical advantage of a fixed pulley is: 1

(a) 2 **(b)** 0 **(c)** less than 1 **(d)** 1

2. The kinetic energy of a body of mass m moving with velocity v is: 1

(a) (1/2) m v **(b)** m v squared

(c) (1/2) m v squared **(d)** m v

3. A body of mass 5 kg is raised through 4 m. The gain in potential energy (g = 10) is: 1

(a) 50 J **(b)** 20 J **(c)** 200 J **(d)** 9 J

4. A movable pulley has a mechanical advantage of: 1

(a) 4 **(b)** 1 **(c)** 1/2 **(d)** 2

5. The gravitational potential energy of a body of mass m at height h is: 1

(a) m g / h **(b)** m h / g **(c)** (1/2) m g h **(d)** m g h

6. A freely falling body: its potential energy ___ and kinetic energy ___. 1

(a) increases, decreases **(b)** decreases, decreases

(c) decreases, increases **(d)** increases, increases

7. An inclined plane helps by: 1

(a) reducing the work done **(b)** reducing the effort needed

(c) increasing the work done **(d)** creating energy

8. A stretched rubber band stores: 1

(a) kinetic energy **(b)** heat

(c) sound energy **(d)** elastic potential energy

9. The total mechanical energy of a freely falling body (no air resistance) is: 1

(a) decreasing **(b)** constant **(c)** zero **(d)** increasing

10. The mechanical advantage of an inclined plane of length l and height h is: 1

(a) l / h **(b)** l - h **(c)** h / l **(d)** l x h

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Define mechanical advantage of a machine. 2

12. Define power. Give the SI unit. 2

13. Define the commercial unit of energy and relate it to the joule. 2

14. A body of mass 10 kg is at a height of 5 m. Find its potential energy (g = 10). 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A ball of mass 0.5 kg is thrown upward with a speed of 10 m/s. Find the maximum height by the conservation of energy (g = 10). 3

16. Differentiate between kinetic and potential energy with two examples each. 3

17. A crane lifts a mass to the 10th floor in time t and then to the 20th floor in 2t. Compare the work done and the power. 3

OR

A pump raises 500 kg of water to a height of 20 m in 100 s. Find the power of the pump (g = 10). 3

18. Why does a perpetual-motion machine not exist? Give two reasons. 3

OR

State three factors that decide the efficiency of a machine and write its formula. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the working of fixed and movable pulleys and block-and-tackle with mechanical advantage values. 5

OR

A body of mass 1 kg falls from a height of 20 m. Find its speed and KE just before impact, and the PE and KE at 5 m above the ground (g = 10). 5

20. Explain the inclined plane as a simple machine. A drum of 100 kg is raised by 1.5 m using a 6 m ramp. Find the force needed (ideal) and the MA (g = 10). 5

OR

Explain why work done by a machine cannot exceed the work done on it, using the example of a lever and calculate the effort for a 100 N load with effort arm 1 m and load arm 0.2 m. 5

END OF QUESTION PAPER Total: 40 Marks

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